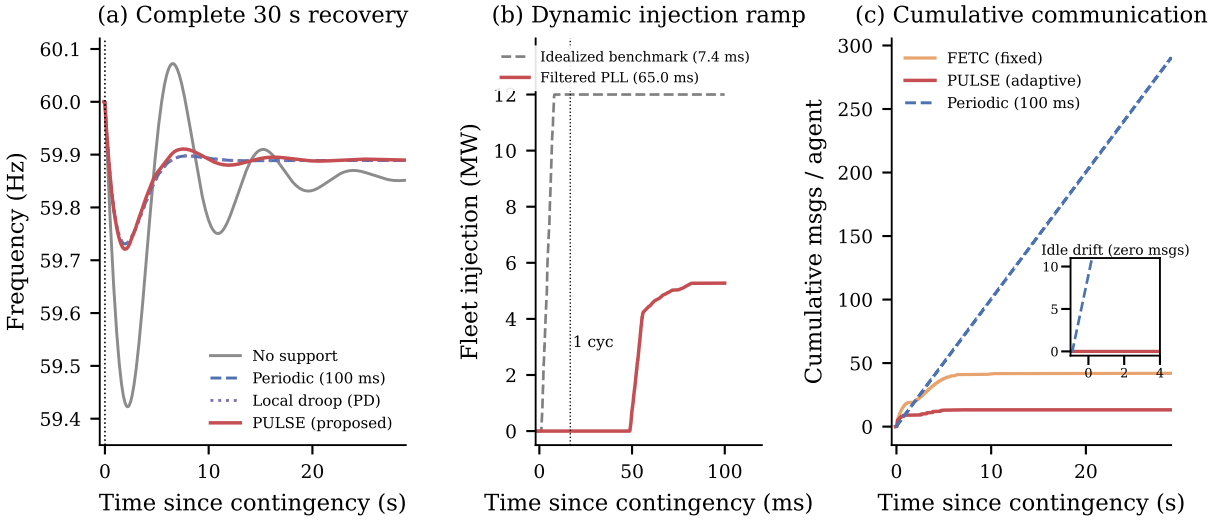
(a) Complete 30 s recovery
Frequency (Hz)
60.1
60.0
59.9
59.8
59.7
59.6
59.5
59.4
0
10
20
Time since contingency (s)
No support
Periodic (100 ms)
Local droop (PD)
PULSE (proposed)
(b) Dynamic injection ramp
Fleet injection (MW)
12
10
8
6
4
2
0
0
50
100
Time since contingency (ms)
Idealized benchmark (7.4 ms)
Filtered PLL (65.0 ms)
1 cyc
(c) Cumulative communication
Cumulative msgs / agent
300
250
200
150
100
50
0
0
10
20
Time since contingency (s)
FETC (fixed)
PULSE (adaptive)
Periodic (100 ms)
Idle drift (zero msgs)
10
5
0
0
2
4